An Internship Report

Of

Zero Trust Cloud Security

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)

By

Student ID: 2410030171
Name of Student: Mukesh
Programme: B.Tech (Computer Science and Engineering)

CANDIDATE'S DECLARATION

I, Mukesh, do hereby declare that the internship report titled “Zero Trust Cloud Security” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: **Mukesh**
Student ID: **2410030171**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and Zscaler for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Zero Trust Cloud Security. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:
Student Name: **Mukesh**

TABLE OF CONTENTS

3. INTERNSHIP COMPLETION CERTIFICATE


शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills®
Nation Building Through Skills

AICTE - EduSkills
2026
VIRTUAL INTERNSHIP

Certificate of Virtual Internship

Mukesh

IILM University, Greater Noida

has successfully completed the 10-weeks

Zero Trust Cloud Security Virtual Internship

During January - March 2026

May this Internship learning propel you toward a bright and successful career.

Supported By zscaler™

Prameet Chhabra
Vice President, Platform Enablement
Zscaler

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

INTERNSHIP GRADE
E

Certificate ID : 4b52beba3f0fefd484c6

Student ID: Mukesh

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 10-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of "Zero Trust Cloud Security". The internship was carried out from January 2026 to March 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The programme was supported by Zscaler, a global leader in cloud security, and was designed to build a strong conceptual and practical foundation in modern Zero Trust security architectures used to protect users, applications, and data across cloud and hybrid environments.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments and weekly assessments. On successful completion of all the requirements, a Virtual Internship Certificate was awarded jointly by EduSkills Academy and Zscaler.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme "Nation Building Through Skills". EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal.

This particular internship was supported by Zscaler, a cloud-native cybersecurity company and a pioneer of the Zero Trust Exchange platform, which secures users, workloads, and devices over the internet, without relying on traditional network perimeters. The collaboration gave learners exposure to industry-grade Zero Trust and Secure Access Service Edge (SASE) concepts.

4.3 Problem Statement

Traditional, perimeter-based security models assume that everything inside an organization's network can be trusted. With the rise of cloud computing, remote work, and Bring-Your-Own-Device (BYOD) practices, this assumption no longer holds — users, applications, and data now live outside the traditional perimeter. The problem addressed by this internship was to understand and apply Zero Trust principles — verifying every user and device, enforcing least-privilege access, and protecting data — regardless of where the request originates.

4.4 Project Objectives

- To understand the core principles of the Zero Trust security model and why it replaces perimeter-based security.
- To learn Identity and Access Management (IAM) concepts including SSO and Multi-Factor Authentication.
- To understand Zero Trust Network Access (ZTNA) as a secure alternative to traditional VPNs.
- To study Secure Access Service Edge (SASE) and the convergence of networking with security.
- To gain awareness of Cloud Access Security Broker (CASB) and Secure Web Gateway (SWG) capabilities.
- To learn Data Loss Prevention (DLP) techniques for protecting sensitive information in the cloud.
- To understand Cloud Security Posture Management (CSPM) for identifying misconfigurations and risks.
- To apply Zero Trust concepts in a capstone project designing a secure cloud access framework.

4.5 Scope of the Project

The scope of this internship covers the foundational and applied aspects of Zero Trust security in cloud environments — including identity verification, device trust, network access control, application-level security, and continuous monitoring. It also introduces the broader SASE framework that unifies networking and security delivery from the cloud. The internship was limited to a virtual, simulation/lab-based learning environment provided by EduSkills Academy and Zscaler, and did not involve implementation on any live organizational production network.

4.6 Technologies and Tools Used

Category	Technologies / Concepts
Security Model	Zero Trust Architecture (ZTA), Least-Privilege Access
Identity & Access	Identity and Access Management (IAM), SSO, Multi-Factor Authentication (MFA)
Network Access	Zero Trust Network Access (ZTNA), Microsegmentation
Cloud Security	Secure Access Service Edge (SASE), Cloud Access Security Broker (CASB)

Web & Data Security	Secure Web Gateway (SWG), Data Loss Prevention (DLP)
Posture Management	Cloud Security Posture Management (CSPM)
Platform	Zscaler Zero Trust Exchange (ZIA / ZPA concepts)

4.7 System Architecture

The general Zero Trust architecture followed during the internship can be summarized in the following layers:

- Identity Layer: Every user and device is authenticated and continuously verified before access is granted.
- Device Trust Layer: Device posture (patch level, compliance, security state) is assessed before granting access.
- Network Access Layer (ZTNA): Access is brokered per-application rather than granting broad network-level access, replacing legacy VPNs.
- Policy & Enforcement Layer: Centralized, context-aware policies determine access based on identity, device, location, and risk.
- Application Layer: Applications are shielded from direct exposure to the internet and are reached only through verified, brokered connections.
- Data Layer: Data Loss Prevention and encryption controls protect sensitive information in transit and at rest.
- Monitoring & Analytics Layer: Continuous logging, threat analytics, and anomaly detection support adaptive, risk-based access decisions.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and weekly assessments, as summarized in the table below.

Week	Module	Description
1	**Foundations of Zero Trust Security**	Introduction to the Zero Trust model, its core principles ("never trust, always verify"), and why perimeter-based security is insufficient for cloud and hybrid work.
2	**Identity & Access Management (IAM)**	Identity as the new security perimeter; authentication, authorization, Single Sign-On (SSO), and Multi-Factor Authentication (MFA).
3	**Zero Trust Network Access (ZTNA)**	Concepts of ZTNA, replacing traditional VPNs with application-level, identity-aware secure access.
4	**Secure Access Service Edge (SASE)**	Convergence of networking and security into a single cloud-delivered service model; SD-WAN and security integration.
5	**Cloud Access Security Broker (CASB)**	Visibility and control over cloud application usage, shadow IT discovery, and enforcement of cloud security policies.
6	**Secure Web Gateway & Data Protection**	Web traffic inspection, threat prevention, and Data Loss Prevention (DLP) techniques for protecting sensitive data.

Week	Module	Description
7	**Cloud Security Posture Management (CSPM)**	Continuous assessment of cloud environments for misconfigurations, compliance gaps, and risk exposure.
8	**Microsegmentation & Least-Privilege Access**	Reducing lateral movement through microsegmentation, least-privilege policies, and continuous access evaluation.
9	**Threat Monitoring & Zero Trust Analytics**	Real-time monitoring, logging, and analytics for detecting anomalies and enforcing adaptive risk-based policies.
10	**Capstone Project & Assessment**	Applying Zero Trust principles to design a secure cloud access framework for an organization, followed by the final assessment.

Each week concluded with a weekly assessment to evaluate conceptual understanding. In the final week, a capstone exercise and final assessment were used to determine the overall internship grade.

4.9 Expected Outcomes

- Clear understanding of the Zero Trust security model and its “never trust, always verify” philosophy.
- Practical awareness of Identity and Access Management, SSO, and Multi-Factor Authentication.
- Understanding of Zero Trust Network Access (ZTNA) as a modern alternative to VPNs.
- Awareness of SASE architecture and the convergence of networking and security.
- Ability to identify cloud misconfigurations using Cloud Security Posture Management concepts.
- Understanding of Data Loss Prevention techniques for protecting sensitive cloud data.
- A completed capstone exercise applying Zero Trust principles to a secure cloud access design.

4.10 Certificate of Completion

The Internship Completion Certificate issued jointly by EduSkills Academy and Zscaler, confirming successful completion of the 10-week AICTE – EduSkills Virtual Internship Program in Zero Trust Cloud Security, is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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5. AICTE – EduSkills National Internship Portal, https://internship.aicte-india.org/